

# AWS checklist

|                |                                                                           |
|----------------|---------------------------------------------------------------------------|
| Station name   | QAS-U                                                                     |
| People present | MRK 295 PELOT instructed by:                                              |
| Purpose        | standard maintenance / installation / take-down / quick visit / emergency |
| Date           | 24/5-2017 13 <sup>45</sup>                                                |

Notes:

| Part                            | Part number upon arrival                                                                                     | Part number upon departure, if replaced | Check after maintenance:                                                            |
|---------------------------------|--------------------------------------------------------------------------------------------------------------|-----------------------------------------|-------------------------------------------------------------------------------------|
| Radiometer                      |                                                                                                              |                                         | - Clean?<br>- Aligned with inclinometer and mast?                                   |
| Inclinometer                    |                                                                                                              |                                         | - Aligned with radiometer and mast?                                                 |
| Satellite antenna               |                                                                                                              |                                         |                                                                                     |
| Wind sensor                     |                                                                                                              |                                         | - Black box pointing true south (i.e. along the mast)?                              |
| Temperature / humidity assembly |                                                                                                              |                                         | - Fan spins and sounds OK?                                                          |
| HygroClip in T/hum assembly     |                                                                                                              |                                         | - NB: could have a 40 °C offset                                                     |
| Sonic ranger on AWS             |                                                                                                              |                                         | - Old sensor with new membrane: Y/N                                                 |
| Solar panel                     |                                                                                                              |                                         | - Clean?<br>- NB: is not effective when partially covered                           |
| GPS antenna                     |                                                                                                              |                                         | - Stuck to top of logger enclosure?                                                 |
| Data logger                     |                                                                                                              |                                         |                                                                                     |
| Multiplexer                     |                                                                                                              |                                         | - Clicking during measurement cycle?                                                |
| Iridium modem IMEI number       |                                                                                                              |                                         | - Transmits?                                                                        |
| Sonic ranger on stakes          |                                                                                                              |                                         | - Old sensor with new membrane: Y/N<br>- NB: make sure not being buried next winter |
| Thermistor string               |                                                                                                              |                                         |                                                                                     |
| Ablation hose assembly          |                                                                                                              |                                         | - No air in hose and bladder half full?                                             |
| Logger enclosure                | - Replaced (including everything inside): Y/N<br>- New desiccant bags: Y/N                                   |                                         |                                                                                     |
| Battery box                     | - New box or new batteries: Y/N<br>- New desiccant bags: Y/N                                                 |                                         |                                                                                     |
| Stakes                          | - New ones drilled in: Y/N<br>- Length of new stakes 8 m: Y/N<br>- Caps in bottom end to reduce melt-in: Y/N |                                         |                                                                                     |

**Metadata BEFORE maintenance** (if establishing a new AWS: skip this)

|                                                                        |                 |
|------------------------------------------------------------------------|-----------------|
| Time difference between logger clock and UTC                           | + / -           |
| Download data to PC and/or change CF Card (wait for green light)       | Y / N           |
| Name of logger program                                                 |                 |
| Screen dump of values in fast scan mode                                | Y / N           |
| Photos of tripod, sensors, logger box wiring etc. (more is better)     | Y / N           |
| Mast tilt in boom direction (+ if radiometer tilting down)             | °               |
| Mast tilt across boom looking from radiometer (- if clockwise)         | °               |
| Boom direction relative to north                                       | ° true/magn     |
| Box on wind sensor exactly towards mast? If not, measure direction.    | 330 Ø / N       |
| Radiometer aligned with mast and inclinometer? If not, measure.        | 2, 290 Ø / N    |
| Temperature / humidity height (bottom of casing – surface)             | 270 cm          |
| Sonic ranger on stakes height (membrane – surface)                     | 280 cm          |
| Free length of stake – outer, holding boom with sonic ranger           | 576 cm          |
| Free length of stake – middle                                          | 546 cm          |
| Free length of stake – outer, holding boom without sonic ranger        | 572 cm          |
| Depth of ablation hose using markings on hose                          | 2520-1030 cm    |
| Height of ablation hose fluid level above the ice surface              | 150-1630 cm     |
| Vertical difference in surface height (at station – at hose) (approx.) | 0 cm            |
| Length of thermistor string on surface from surface marking            | 220 cm          |
| Stake assembly still standing?                                         | Y / N / sort of |
| Apparent damage to sensors? If yes, which?                             | Y / N           |

**Metadata AFTER maintenance / establishment**

|                                                                          |             |
|--------------------------------------------------------------------------|-------------|
| Time logger clock set to UTC?                                            | Y / N       |
| Name of logger program                                                   |             |
| Screen dump of values in fast scan mode                                  | Y / N       |
| Latitude/longitude (if new or relocated)                                 |             |
| Photos of tripod, sensors, logger box wiring etc. (more is better)       | Y / N       |
| Mast tilt in boom direction (+ if radiometer tilting down)               | °           |
| Mast tilt across boom looking from radiometer (- if clockwise)           | °           |
| Boom direction relative to north (radiometer should point to true south) | ° true/magn |
| Temperature / humidity height (bottom of casing – surface)               | cm          |
| Sonic ranger on stakes height (membrane – surface)                       | cm          |
| Free length of stake – outer, holding boom with sonic ranger             | cm          |
| Free length of stake – middle                                            | cm          |
| Free length of stake – outer, holding boom without sonic ranger          | cm          |
| Depth of ablation hose using markings on hose                            | cm          |
| Height of ablation hose fluid level above the ice surface                | cm          |
| Vertical difference in surface height (at station – at hose) (approx.)   | cm          |
| Length of thermistor string on surface from surface marking              | cm          |